

Determination of Technical Support Requirements of an Office Automation System

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Abstract

The Body & Chassis Engineering Department of Ford Motor Company recently formulated plans for greatly increasing (from 155 to 476) the number of Fortune engineering-workstation computers. This increase represents a vital step in the implementation of an office automation system. Systems-analysis personnel of the Department used discrete simulation as a tool to quantify the impact of this increase on the workload of technical support personnel. In turn, Department management used the results of these simulations to formulate a hiring plan consistent with the planned rate of terminal acquisition.

Introduction

In February, 1985, systems-analysis personnel of the Body & Chassis Engineering Department requested simulation support from the Simulation and Real-Time Systems Section of the Computer Aided Engineering Systems Department. The systems analysts described the current operational milieu within their office and the planned threefold increase in the number of Fortune workstations. The need for simulation arose from the lead time involved in the hiring of any additional support personnel. That is to say, if hiring decisions were deferred until an inadequacy of support level manifested itself in real time, an unwanted interval of inadequate workstation support would inevitably occur while a hiring decision was implemented.

Development of the Model

The General Purpose Simulation System (GPSS) language was chosen for model development because of its ready availability and the rapidity with which a model could be coded, verified, and validated. Ford Motor Company has developed a GPSS pre- and post-processor, the General Transfer Line Emulator (GENTLE). Given operational parameters of various types of canonical production facilities, GENTLE can automate the coding of a GPSS model [5]. This GPSS model, however, was more efficiently coded directly because of its small size (162 blocks), use of many priority levels, and use of a user chain. The Body and Chassis Engineering systems analysts provided logbooks of technical support workload experience over a 24-workday period. These logbooks specified the interarrival-time distributions of the twelve types of problems for whose resolution the

technical support personnel were responsible. The model comprised a heavy "rush hour" each midmorning and a less pronounced "rush hour" each midafternoon. Also, the model implemented arrival rates varying by the day of the week, inasmuch as Mondays and Tuesdays were characteristically busier than the remaining weekdays. Likewise, the logbooks specified the service-time distributions needed for resolution of each type of problem. Examination of histograms and Kolmogorov-Smirnov goodness-of-fit test results on these data resulted in use of exponential distributions for service times and uniform distributions for interarrival times. The hierarchy of eight classes used in day-to-day operational procedure for prioritizing arriving service requests was also incorporated into the model. Furthermore, the model reflected the necessity of each technician's devoting one hour daily to administrative work. In these contexts, Simulation and Real-Time Systems built a GPSS model using the current technical staffing level and current number of workstations (155), as reflected in problem interarrival rates. The model, using a clock-tick of ten seconds, ran for a simulated 90 work-days of nine hours each.

Construction of this GPSS model proceeded smoothly, requiring 25 hours of programming-development time. The model contained an example of the flexibility and convenience of GPSS user chains. Arriving problems, represented as transactions, had their priority loaded into a parameter; then, those transactions were LINKed onto a user chain according to the value of that transaction parameter. In GPSS, larger integers matched higher priority; therefore, lower-priority problems were linked nearer the front end of the user chain. At the UNLINK block, upon availability of a technician, transactions were unlinked from the rear of the user chain. GPSS routinely provided the average time spent by transactions on this user chain, plus the average and maximum lengths of the chain. Each of these quantities represented a performance measure of high interest to the systems analysts. Additionally, since transactions on user chains need not be examined on each update of the simulation clock, welcome run-time efficiencies accrued.

Verification and Validation of the Model

The model was verified without undue difficulty. The model coder presented the GPSS code in structured walkthroughs to two other programmers experienced in GPSS. Examinations of block counts (a valuable debugging aid the author thinks too little used) and GPSS statistical output were additional important steps to the goal of assurance the model was performing exactly as intended.

As an important first step in model validation, the systems analysts in Body and Chassis Engineering were vigorously urged to read all comments in the GPSS model. Inasmuch as these comments incorporated the modeler's understandings of system operation, any misunderstandings would very likely reveal themselves in incorrect comments. Indeed, the systems analysts did thereby expose an error in the way technicians scheduled their daily hour of administrative work. Once laid bare, this error was easily corrected. Next, model outputs (which incorporated the current technician-staffing level and the current problem-arrival rates) were compared with actual observations of systems performance (e.g., average number of busy technicians and average number of problems queued awaiting

attention). All such compared observations agreed satisfactorily, with a maximum disparity of 4%. Next, the question of simulation run length was examined. Conceptually, since the system started from empty each morning, and since the arrival rates, as a function of the weekday, recycled each Monday, the simulation was terminating as opposed to steady-state. The sequential procedure described in Law and Kelton [2](8.5.2) was used to evaluate confidence intervals for system performance measures in terms of simulation run length. Run lengths of 90 work-days (18 work-weeks) produced results significant at the 10% level (90% confidence intervals). Uncertainties at this confidence level were small compared to inevitable fluctuations in technician support level due to illnesses and vacations. Therefore, the model earned managerial confidence as a firm base for staffing-level decisions. The model was then accepted as valid. This acceptance occurred three weeks after the initial request for simulation support.

Use and Extension of the Validated Model

The model's parameters were changed in two ways concurrently. First, redefinition of the STORAGE named "OPERS," changing its capacity, simulated the effects of potential hiring. Second, changing the means and half-ranges of the uniform interarrival-time distributions simulated increases in the number of workstations. The mean of the interarrival-time distribution was assumed inversely proportional to the number of workstations; e.g., to simulate doubling the number of workstations, the interarrival time was halved. The variance of the interarrival time was assumed inversely proportional to the square root of the number of workstations. Problem-resolution times were assumed independent of both the number of workstations and the number of technicians. These changes affected numbers in, but not the logical structure of, the model. Hence, a variety of scenarios relative to the numbers of workstations and technicians could be simulated quickly. Examination of such scenarios usually revealed a sharp increase in system performance at a certain level of technician support. For example,

405 Workstations

Number of technicians	Average problem wait time (minutes)	Average waiting queue
3	84	7.1
4	8	0.7

Hence, management was readily able to specify staffing levels for each planned increase in the number of workstations.

Conclusions

The Body & Chassis Engineering workstation-acquisition plans are now two-thirds complete. Staffing increases have been implemented as suggested by runs of this model. Model predictions have agreed well with ongoing observations of system performance. Maximum disparities in average problem wait time and average queue length have been 8%. Most of these disparities have been attributed to longer problem-resolution times

occurring when a problem is assigned to a new, as yet incompletely trained technician. The assumption of independence of problem-resolution times and the number of technicians will gradually become accurate as the new technicians' training is completed. Important contributors to the success of this simulation were the availability of statistical expertise in both departments involved and the high level of user participation throughout model development.

Postscript

In September, 1986, systems-analysis personnel of the Body & Chassis Engineering Department requested simulation support from the Simulation and Real-Time Systems Section pertinent to assessing plans for expansion of computer-graphics hardware, usage, and support. This request for simulation in a new, but similar, context indicates that simulation has demonstrated its valuable role in managerial planning.

Acknowledgements

The author thanks Professor Onur M. Ülgen, Department of Industrial & Systems Engineering, University of Michigan - Dearborn and Roy A. Asher, colleague in the Simulation and Real-Time Systems Section, for valuable advice on technical content and clarity.

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